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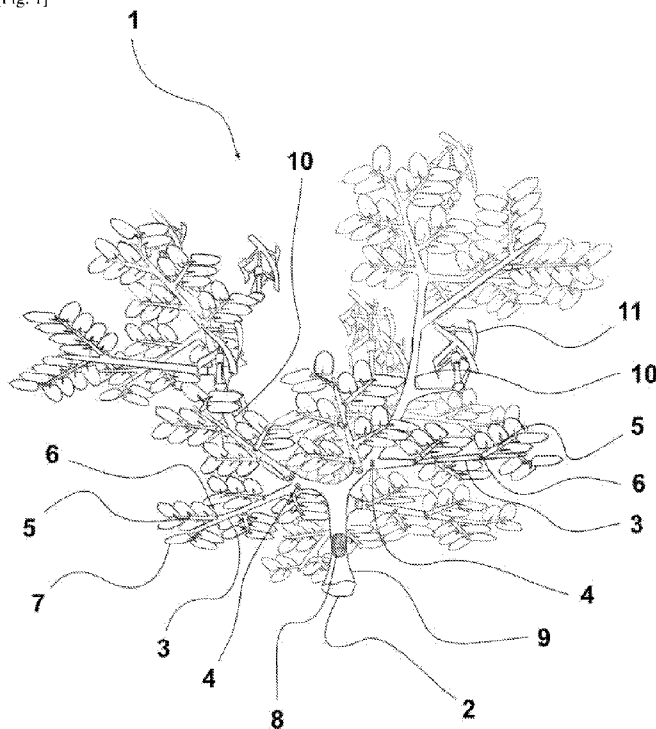
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(54) Title: A SOLAR TREE

[Fig. 1]



(57) Abstract: The invention relates to a solar tree (1) comprising: a trunk (2); at least one first branch (3) rotatably connected to the trunk (2) and driven by a first motor (4); at least one second branch (5) rotatably connected to the first branch (3) and driven by a second motor (6); and at least two photovoltaic panels (7) fixedly connected to the second branch (5), wherein the photovoltaic panels (7) are connected in parallel to each other and configured to charge a battery (8).

Declarations under Rule 4.17:

- *as to the identity of the inventor (Rule 4.17(i))*
- *as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))*
- *as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))*

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- *in black and white; the international application as filed contained color or greyscale and is available for download from PATENTSCOPE*

Description

Title of Invention: A solar tree

Technical Field

[0001] This invention relates to artificial tree structures fitted with leaf-shaped photovoltaic panels or other means for generating renewable electric energy.

Background Art

[0002] In prior art, artificial tree structures are known to be provided with means for producing renewable electric energy, such as photovoltaic panels in the shape of leaves or wind power plants. The term "tree" in the context of the present invention refers to a tree-like or bush-like columnar structure.

[0003] The Chinese patent application CN 113872497 A and the Chinese utility model CN 212324025 U disclose (see [Fig.2] and corresponding reference signs) a solar tree comprising: a base 1; a trunk 2 fixed on the base 1; at least one first branch 3 rotatably connected to the trunk 2 and driven by a first servomotor 4; at least one second branch (no reference sign) fixedly connected to the first branch 3; one photovoltaic panel 5 rotatably connected to each second branch and driven by a second servomotor 6. The effect is the tilting of all photovoltaic panels 5 on one first branch together (when the second servomotors 6 are idle) or of each photovoltaic panel 5 separately (when the second servomotors 6 are in operation). A disadvantage of this solar tree is that if one panel is shaded, that panel must be rotated by a separate second servomotor to a better position relative to the sun. However, a shaded panel requiring rotation or tilting already produces no (or only minimal) electric energy even when partially shaded, so the energy produced by other panels must be used to rotate the corresponding second servomotor. If shading occurs on multiple panels at the same time, or on almost all of them, then there are not enough unshaded panels on the tree relative to the second servomotors. The ratio of unshaded panels to second servomotors is always a maximum of 1:1 at 100 % illumination, and at partial shading there are fewer unshaded panels than second servomotors (e.g., 1:2, 1:3, etc.). Thus, the energy consumption for rotating each shaded panel is high since many to all of the second servomotors must be in operation when searching for a more suitable position relative to the sun in such multiple shading. The solar tree itself thus has to consume more energy from elsewhere to operate, e.g. energy stored in a battery or from the grid.

[0004] The Chinese patent application CN 113872498 A discloses (see Figures 1, 2 and corresponding reference signs) a solar tree comprising: a base 1; a trunk 3 rotatably connected to the base 1 and driven by a first servomotor 2; at least one first branch 4 fixedly connected to the trunk 3; at least one second branch (no reference sign) fixedly

connected to the first branch 4; one photovoltaic panel 5 rotatably connected to each second branch and driven by a second servomotor 6. The effect is the tilting of all photovoltaic panels 5 on all first branches together (when the second servomotors 6 are idle) or of each photovoltaic panel 5 separately (when the second servomotors 6 are in operation). A disadvantage of this solar tree is the low variability of panel rotation within the different first branches, as all first branches are fixedly connected to the trunk.

[0005] The Chinese patent application CN 104779895 A discloses a solar-wind tree comprising: a trunk; at least a first branch fixedly connected to the trunk and provided with one photovoltaic panel and with one wind power plant. The disadvantage of this solar tree is the lack of ability to rotate the branches or panels in case of shading.

[0006] The Korean patent application KR 20120023961 A discloses a solar tree with a branch and photovoltaic panels attached to it. The branch is divided into a plurality of vertically rotatable segments, wherein at least one photovoltaic panel is fixedly connected to each segment, and wherein the segments can rotate relative to each other. A disadvantage of this solar tree is that the branches or panels can only be rotated in the vertical axis in case of shading, not in all degrees of freedom.

[0007] The Chinese utility model application CN 207633840 U discloses (see Figures 1, 2 and corresponding reference signs) a solar tree comprising: a pipe body 1; at least one trunk 122 rotatably connected to the pipe body 1 and driven by a controller for controlling a rotating shaft 11; at least one branch 124 fixedly connected to the trunk 122; and at least two photovoltaic panels 121 fixedly connected to the branch 124 and configured to charge a battery arranged on a platform 3. A disadvantage of this solar tree is the lack of ability to rotate the branches or panels in case of shading.

[0008] Thus, in the prior art, there emerges a need and the object of the present invention is to provide a solar tree exhibiting lower power consumption for self-manipulation in case of shading.

Summary of Invention

[0009] The object of the present invention is achieved by the solar tree according to claim 1. Preferred embodiments of the solar tree are according to claims 2 to 14.

[0010] The underlying idea of the solar tree in the embodiment with a rotatable connection of the first and second branches according to the present invention is that in case of partial shading of one or more panels, but not all panels, within one second branch, the parallel connection of the panels ensures that sufficient electrical power is produced by the remaining fully illuminated panels to recharge a battery (integral or external) and to drive the corresponding second motor to rotate or tilt the entire second branch from the battery, thereby reducing shading. While the battery power is consumed to rotate or tilt

the entire second branch, at the same time the consumption is balanced by the output of the remaining unshaded panels. If shading occurs on multiple panels at the same time, or on almost all of them, there are a sufficient number of unshaded panels on the tree relative to the second motors. The ratio of unshaded panels to second motors is always more than 1:1 at 100 % illumination (e.g., 2:1, 3:1, 4:1, 5:1), etc., and at partial shading there may still be more unshaded panels than second motors (e.g., 2:1, 3:1). Thus, the energy consumption to rotate each shaded panel is lower despite the fact that many to all of the second motors have to be in operation when searching for a more suitable position relative to the sun in such multiple shading. However, the consumption of these second motors is offset by the energy production of the remaining panels. The solar tree itself thus uses less energy from elsewhere to operate, e.g. energy stored in the battery or from the grid.

- [0011] The second branches with photovoltaic panels do not have to be attached to the solar tree facing south, because thanks to the rotation mechanism they can always be rotated perpendicular to the sun even if the second branch is facing north.
- [0012] Another advantage of multiple panels on one second branch is the possibility to turn their active surface downwards quickly and simultaneously in the event of hail (for protection) or maintenance (for cleaning with a stream of pressurized water from below).
- [0013] Similar to the prior art, the tilting of all photovoltaic panels on one first branch is common (when the second motors are idle) or individual for each photovoltaic panel (when the second motors are active).
- [0014] Preferably, the first and second motors are connected to a battery, preferably arranged in the trunk (but other locations are also available, e.g. underground or in the branches). The battery is used to store the electric energy produced and can be further used for the operation of the solar tree itself (e.g. for tilting) as well as for other applications and functions of the solar tree (see below).
- [0015] Preferably, the photovoltaic panels are configured to drive the second motor by connecting via a battery (i.e. first storing the energy in the battery and then using it to drive the second motor).
- [0016] Preferably the solar tree is connected to an additional external battery for use in households, commercial premises or industry. The solar tree thus allows energy saving in applications outside the tree.
- [0017] Preferably the motor is a servomotor.
- [0018] Preferably, the photovoltaic panel is bifacial and includes a reflective foil to extract energy from the other (bottom) side of the panel.
- [0019] Preferably, the trunk is divided into a central trunk portion and at least one branched trunk portion, wherein at least one first branch is rotatably connected to the branched

trunk portion. The division allows for greater geometric variation in the arrangement of branches and panels in space. In the context of the present invention as a whole, the trunk is at least a central trunk portion.

[0020] Preferably, the solar tree further comprises a wind power plant fixedly connected to the trunk, in particular to the branched trunk portion. By wind power plant, in the context of the present invention, a small wind turbine or a small vertical wind power plant is understood. In this embodiment, the possibility of controlling the position and rotation of the panels is further advantageous, since the wind power plant itself can also be a source of shading for the panels. The wind power plants are always on a separate branch for better use of the wind energy.

[0021] In particular, the above-described wind power plant can be of Savonius wind turbine type. The Savonius wind turbine comprises two wind turbine blades which are symmetrically arched, attached to a vertical axis body and arranged opposing to each other. The vertical axis body is rotatably anchored between a lower support element and an upper support element, preferably by means of bearings. The support elements further hold a plurality of directing blades therebetween. The directing blades are therefore stationary and can have a quasi-rectangular shape with at least one of the long sides being arched, for example along a spherical or paraboloid curve. The directing blades envelop the wind turbine blades and are oriented at an angle of 30 to 60° with respect to the vertical axis body. Therefore, the directing blades are oriented with respect to the wind turbine blades such that they direct an incoming stream of wind to a greater extent towards a windward (convex) side of one of the wind turbine blades and to a lesser extent away from a leeward (concave) side of the other of the wind turbine blades, therefore avoiding slowing down the rotation of the wind turbine blades. The lower support element can also comprise an outlet element (e. g. a channel or an opening)) which allows the wind to escape the space of the wind turbine blades once it pushes the wind turbine blades to rotate. There can be a generator arranged below the lower support element, generally comprising a rotor and a stator. The rotor is connected with the rotatable vertical axis body and the stator is connected with the remainder of the solar tree. The Savonius wind turbine is suitable for areas with low-intensity winds. The wind turbine blades, the directing blades, the vertical axis body and the support elements can be made from a transparent polymer, such as poly(methyl)methacrylate (PMMA), to avoid creating shade on the photovoltaic panels.

[0022] Preferably, an illumination sensor is arranged at a distal end of the first or second branch or of the branched trunk portion for sensing the current angle of illumination of specific parts of the solar tree by solar radiation.

- [0023] Preferably, the solar tree further comprises a charging station arranged e.g. in the trunk, serving as a charging post for e.g. electric bicycles or electric cars.
- [0024] Preferably, the solar tree further comprises a control unit connected to the photovoltaic panels, the battery, and optionally to the illumination sensor, the wind power plant and the charging station. The control unit continuously evaluates the power output of the individual panels or wind power plants and coordinates the rotation of the individual motors to ensure the highest possible illumination of the panels. The possibility of remote control of the rotation of the panels is also foreseen.
- [0025] Preferably, the trunk (i.e. the central trunk portion, and possibly the branched trunk portions) is formed as a steel skeleton in combination with a concrete shell, and the first and second branches are formed from aluminium and/or an aluminium alloy. The first and second branches may be formed as an aluminium tube wrapped on the outside with a fibreglass coating imitating the surface of the tree, which will reduce the manufacturing costs.
- [0026] Preferably, in the solar tree embodiment, where the second branch is rotatably connected to the first branch and driven by the second motor, a pin protrudes from the second branch, on which pin a protruding portion of the first branch is coaxially and rotatably mounted. The protruding portion of the first branch means an open projection arranged on the surface of the first branch. The motor, which is mechanically connected to a gearbox mounted in the protruding portion of the first branch, is arranged inside the protruding portion of the first branch. The gearbox and the pin are mechanically connected by a coupling element for transferring torque from the motor and the gearbox to the pin and the second branch. Electrical cables for powering the photovoltaic panels and transmitting the generated electrical energy for storage are routed through the interior of the protruding portion of the first branch, outside the motor, the gearbox and the coupling element, then through holes in the pin, inside the pin, and then through the interior of the second branch.
- [0027] Another aspect of this invention is the above-described Savonius wind turbine on its own, without being fixedly connected to the trunk, in particular to the branched trunk portion, of the solar tree, and comprising: two wind turbine blades; a vertical axis body rotatably anchored between a lower support element and an upper support element and holding the wind turbine blades; and a plurality of directing blades held between the support elements. The two wind turbine blades are symmetrically arched, attached to the vertical axis body and arranged opposing to each other. The vertical axis body can be rotatably anchored between a lower support element and an upper support element by means of bearings. The directing blades are therefore stationary and can have a quasi-rectangular shape with at least one of the long sides being arched, for example along a spherical or paraboloid curve. The directing blades envelop

the wind turbine blades and are oriented at an angle of 30 to 60° with respect to the vertical axis body. Therefore, the directing blades are oriented with respect to the wind turbine blades such that they direct an incoming stream of wind to a greater extent towards a windward (convex) side of one of the wind turbine blades and to a lesser extent away from a leeward (concave) side of the other of the wind turbine blades, therefore avoiding slowing down the rotation of the wind turbine blades. The lower support element can also comprise an outlet element (e. g. a channel or an opening)) which allows the wind to escape the space of the wind turbine blades once it pushes the wind turbine blades to rotate. There can be a generator arranged below the lower support element. The wind turbine blades, the directing blades, the vertical axis body and the support elements can be made from a transparent polymer, such poly(methyl)methacrylate (PMMA).

Brief Description of Drawings

Fig.1

[0028] [Fig.1] shows a solar tree according to the present invention.

Fig.2

[0029] [Fig.2] shows an exploded view of the rotatable connection of the first and second branches.

Fig.3

[0030] [Fig.3] shows a longitudinal section view of the rotatable connection of the first and second branches.

Fig.4

[0031] [Fig.4] shows a further embodiment of a solar tree according to the present invention, comprising a wind power plant of Savonius wind turbine type.

Fig.5

[0032] [Fig.5] shows a Savonius wind turbine of [Fig.4] in a perspective view.

Fig.6

[0033] [Fig.6] shows a Savonius wind turbine of [Fig.4] in a cross-sectional view.

Examples

[0034] Example 1

[0035] The first example embodiment is a solar tree **1** shown in [Fig.1]. The solar tree **1** comprises: a trunk **2** comprising a central trunk portion **9** and branching trunk portions **10**; a first branch **3**, wherein each first branch **3** is rotatably connected to the trunk **2** (in particular to the branching trunk portions **10**) and driven by a first servomotor **4**; a second branch **5**, wherein each second branch **5** is rotatably connected to the first branch **3** and driven by a second servomotor **6**; and photovoltaic panels **7** fixedly connected to the second branch **5**. These photovoltaic panels **7** are connected in

parallel to each other and are configured to charge an integral battery 8 or an external battery, and subsequently to drive the second servomotor 6. Vertical wind power plants 11 are fixedly connected to the trunk 2, in particular to the branching trunk portions 10, and/or to the first branches 3. The trunk 2 is formed as a steel skeleton in combination with a concrete shell and the first and second branches 3, 5 are formed of aluminium, for example as an aluminium casting or a prefabricated aluminium tube.

[0036] Example 2

[0037] The second example embodiment is the solar tree 1 of Example 1, but without vertical wind power plants due to its location in a place with low wind intensity. In this example, the number of first and second branches 3, 5 can be increased compared to Example 1 to increase the output of the solar tree.

[0038] Example 3

[0039] The third example embodiment is the solar tree 1 according to Example 1 or 2, wherein the first and second branches 3, 5 are formed as a prefabricated aluminium tube wrapped on the outside with a fiberglass surface imitating a tree surface.

[0040] Example 4

[0041] The fourth example embodiment is the solar tree 1 according to any one of Examples 1 to 3, as shown in Figures 4 to 6, wherein the vertical wind power plants 11 are of Savonius wind turbine type. The Savonius wind turbine comprises two wind turbine blades 24 which are symmetrically arched, attached to a vertical axis body 25 and arranged opposing to each other. The vertical axis body 25 is rotatably anchored between a lower support element 21 and an upper support element 23 by means of bearings 27. The support elements 21, 23 further hold a plurality of directing blades 22 therebetween (such as 18 directing blades 22). The directing blades 22 are therefore stationary and have a quasi-rectangular shape with at least one of the long sides being arched, for example along a spherical or paraboloid curve. The directing blades 22 envelop the wind turbine blades 24 and are oriented at an angle of 30 to 60° (such as 30°, 40°, 45°, 50°, 60°) with respect to the vertical axis body 25. Therefore, the directing blades 22 are oriented with respect to the wind turbine blades 24 such that they direct an incoming stream of wind (direction A1 on [Fig.6]) to a greater extent towards a windward (convex) side of one of the wind turbine blades 24 (direction A2 on [Fig.6]) and to a lesser extent away from a leeward (concave) side of the other of the wind turbine blades 24 (direction A3 on [Fig.6]), therefore avoiding slowing down the rotation of the wind turbine blades 24. The lower support element 21 also comprises an outlet element 26 (e. g. a channel or an opening) which allows the wind to escape the space of the wind turbine blades 24 once it pushes the wind turbine blades 24 to rotate. There is a generator 28 arranged below the lower support element

21, generally comprising a rotor and a stator. The rotor is connected with the rotatable vertical axis body 25 and the stator is connected with the remainder of the solar tree 1.

[0042] The above-described Savonius wind turbine is suitable for areas with low-intensity winds. The wind turbine blades 24, the directing blades 22, the vertical axis body 25 and the support elements 21, 23 can be made from a transparent polymer, such poly(methyl)methacrylate (PMMA), to avoid creating shade on the photovoltaic panels 7.

[0043] Example 5

[0044] The fifth example embodiment is the rotatable connection of the first and second branches 3, 5 in the solar tree 1 according to any of Examples 1 to 4, shown in Figures 2 and 3. A pin 12 protrudes from the second branch 5, on which the protruding portion of the first branch 3 is coaxially and rotatably mounted, wherein a sealing ring 17 and bearings 16 are arranged between the pin 12 and the protruding portion of the first branch 3. A motor 6, such as a servomotor, is arranged inside the protruding portion of the first branch 3 and mechanically connected to a gearbox 14 mounted in the protruding portion of the first branch 3 by means of a bracket 15. The gearbox 14 and the pin 12 are mechanically connected by a coupling element 13 for transmitting the torque from the motor 6 and the gearbox 14 to the pin 12, and further to the second branch 5. Electrical cables 19 are routed through the interior of the protruding part of the first branch 3, outside the motor 6, the gearbox 14 and the coupling element 13, further through holes 18 in the pin 12, inside the pin 12, and then through the interior of the second branch 5.

Industrial Applicability

[0045] The solar tree described above can be used, for example, as a design solution for energy saving or as an island system for charging stations for electric bicycles or electric cars.

Reference Signs List

[0046]
 [0047] 1 solar tree
 [0048] 2 trunk
 [0049] 3 first branch
 [0050] 4 first motor
 [0051] 5 second branch
 [0052] 6 second motor
 [0053] 7 photovoltaic panel
 [0054] 8 battery
 [0055] 9 central trunk portion

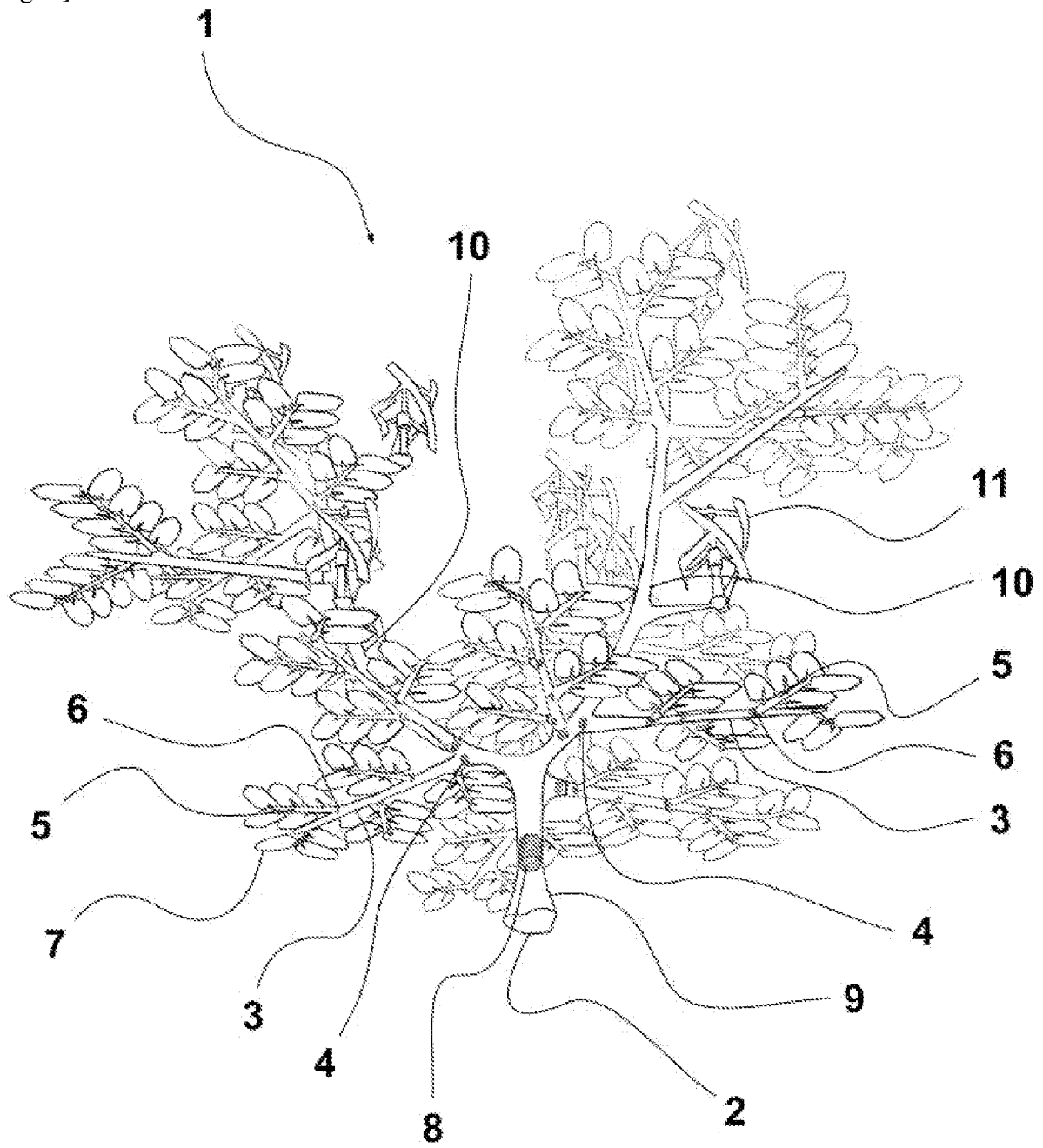
- [0056] 10 branched trunk portion
- [0057] 11 wind power plant
- [0058] 12 pin of the second branch 5
- [0059] 13 coupling element
- [0060] 14 gearbox
- [0061] 15 bracket for the motor 6
- [0062] 16 bearings
- [0063] 17 sealing ring
- [0064] 18 hole in the pin 12
- [0065] 19 cable
- [0066] 20 rotatable connector
- [0067] 21 lower support element
- [0068] 22 directing blade
- [0069] 23 upper support element
- [0070] 24 wind turbine blade
- [0071] 25 vertical axis body
- [0072] 26 outlet element
- [0073] 27 bearing
- [0074] 28 generator
- [0075] A1 direction of incoming wind
- [0076] A2 direction of incoming wind directed towards a windward side
- [0077] A3 direction of incoming wind directed away from a leeward side

Claims

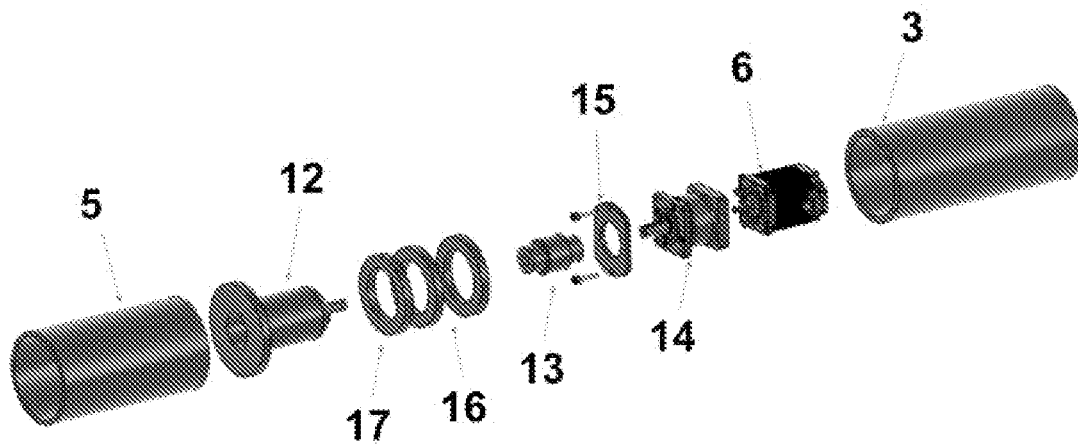
- [Claim 1] A solar tree (1), comprising:
- a. a trunk (2);
 - b. at least one first branch (3) rotatably connected to the trunk (2) and driven by a first motor (4);
 - c. at least one second branch (5) connected to the first branch (3);
- characterised in that** the second branch (5) is rotatably connected to the first branch (3) and driven by a second motor (6),
- and **in that**, the solar tree (1) comprises:
- at least two photovoltaic panels (7) fixedly connected to the second branch (5), wherein the photovoltaic panels (7) are connected in parallel to each other and are configured to charge a battery (8).
- [Claim 2] The solar tree (1) according to claim 1, **characterised in that** the first and second motors (4, 6) are connected to the battery (8), preferably arranged in the trunk (2).
- [Claim 3] The solar tree (1) according to claim 2, **characterised in that** the photovoltaic panels (7) are configured to drive the second motor (6) by connection via the battery (8).
- [Claim 4] The solar tree (1) according to any one of the preceding claims, **characterised in that** it is connected to an additional external battery.
- [Claim 5] The solar tree (1) according to any one of the preceding claims, **characterised in that** the motor (4, 6) is a servomotor.
- [Claim 6] The solar tree (1) according to any one of the preceding claims, **characterised in that** the photovoltaic panel (7) is bifacial.
- [Claim 7] The solar tree (1) according to any one of the preceding claims, **characterised in that** the trunk (2) is divided into a central trunk portion (9) and at least one branched trunk portion (10), wherein at least one first branch (3) is rotatably connected to the branched trunk portion (10).
- [Claim 8] The solar tree (1) according to any one of the preceding claims, **characterised in that** it further comprises a wind power plant (11) fixedly connected to the trunk (2), in particular to the branched trunk portion (10) or to the first branch (3).
- [Claim 9] The solar tree (1) according to any one of the preceding claims, **characterised in that** an illumination sensor is arranged at a distal end of the first or second branch (3, 5) or of the branched trunk portion (10).

- [Claim 10] The solar tree (1) according to any one of the preceding claims, **characterised in that** it further comprises a charging station arranged in the trunk (2).
- [Claim 11] The solar tree (1) according to any one of the preceding claims, **characterised in that** it further comprises a control unit connected with the photovoltaic panels (7), the battery (8), optionally with the illumination sensor, a wind turbine (11) and the charging station.
- [Claim 12] The solar tree (1) according to any one of the preceding claims, **characterised in that** the trunk (2) is formed as a steel skeleton in combination with a concrete shell and the first and second branches (3, 5) are formed of aluminium and/or aluminium alloy.
- [Claim 13] The solar tree (1) according to claim 12, **characterised in that** the first and second branches (3, 5) are formed as an aluminium tube wrapped on the outside with a fiberglass surface.
- [Claim 14] The solar tree (1) according to any one of the preceding claims, wherein the second branch (5) is rotatably connected to the first branch (3) and driven by the second motor (6), **characterised in that** a pin (12) protrudes from the second branch (5), on which pin (12) a protruding portion of the first branch (3) is coaxially and rotatably mounted, wherein the motor (6), which is mechanically connected to a gearbox (14) mounted in the protruding portion of the first branch (3), is arranged inside the protruding part of the first branch (3), wherein the gearbox (14) and the pin (12) are mechanically connected by a coupling element (13) for transmitting torque from the motor (6) and the gearbox (14) to the pin (12) and the second branch (5), wherein electrical cables (19) are routed through the interior of the protruding portion of the first branch (3), outside the motor (6), the gearbox (14) and the coupling element (13), then through holes (18) in the pin (12), inside the pin (12), and then through the interior of the second branch (5).

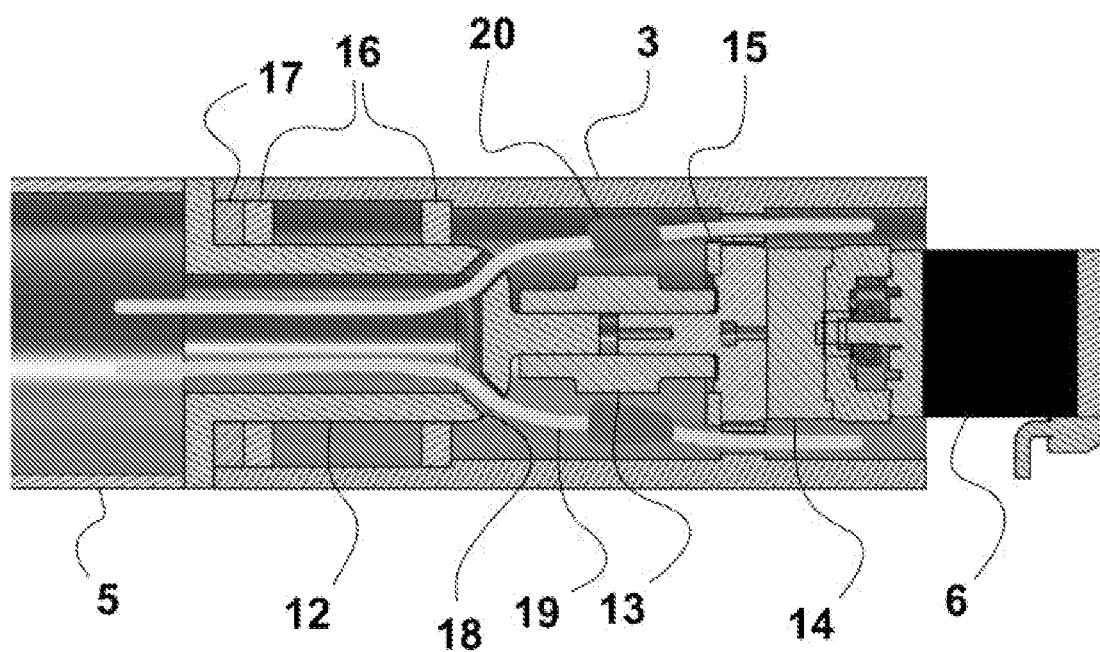
[Fig. 1]



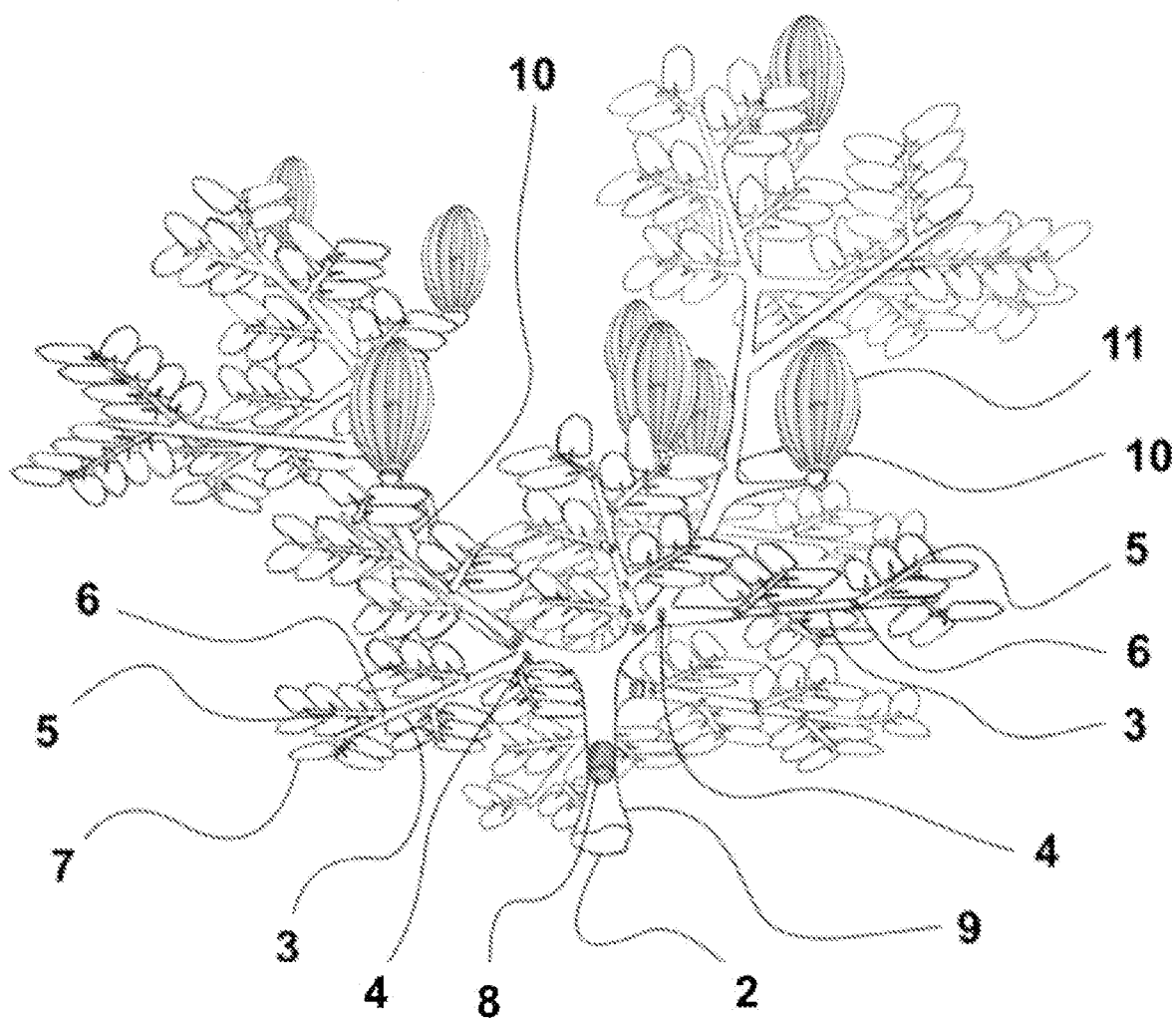
[Fig. 2]



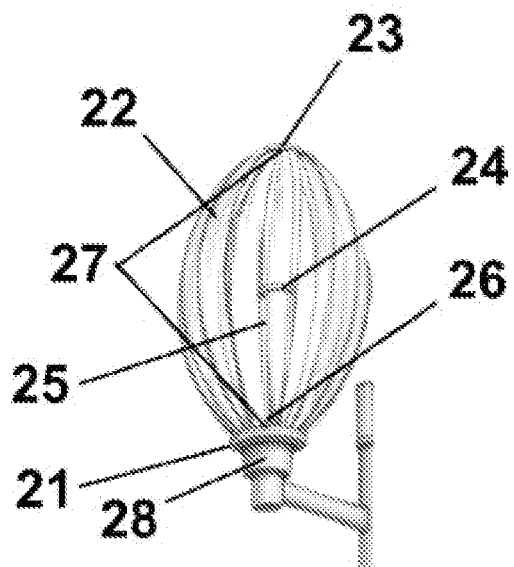
[Fig. 3]



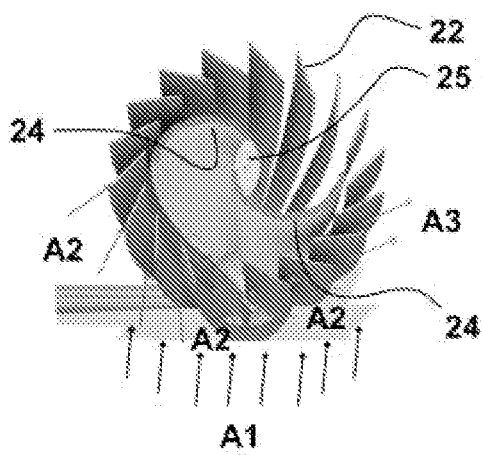
[Fig. 4]



[Fig. 5]



[Fig. 6]



INTERNATIONAL SEARCH REPORT

International application No

PCT/CZ2025/050053

A. CLASSIFICATION OF SUBJECT MATTER

INV. H02S20/32 H02S10/12

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H02S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal**C. DOCUMENTS CONSIDERED TO BE RELEVANT**

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	CN 207 633 840 U (ZHEJIANG KAIYUAN MECH & ELECTRICAL GROUP CO LTD) 20 July 2018 (2018-07-20) cited in the application the whole document -----	1 - 14
A	CN 113 872 497 A (UNIV SHANDONG JIAOTONG) 31 December 2021 (2021-12-31) cited in the application the whole document -----	1 - 14
A	CN 212 324 025 U (UNIV SHANDONG JIAOTONG) 8 January 2021 (2021-01-08) cited in the application the whole document ----- - / - -	1 - 14



Further documents are listed in the continuation of Box C.



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Date of the actual completion of the international search

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INTERNATIONAL SEARCH REPORT

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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	CN 113 872 498 A (UNIV SHANDONG JIAOTONG) 31 December 2021 (2021-12-31) cited in the application the whole document -----	1 - 14
A	CN 104 779 895 A (UNIV GUIZHOU) 15 July 2015 (2015-07-15) cited in the application the whole document -----	1 - 14
A	KR 2012 0023961 A (PARK SO YUN [KR]) 14 March 2012 (2012-03-14) cited in the application the whole document -----	1 - 14

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/CZ2025/050053

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
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CN 113872497	A	31-12-2021	NONE
CN 212324025	U	08-01-2021	NONE
CN 113872498	A	31-12-2021	NONE
CN 104779895	A	15-07-2015	NONE
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